$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

$$\frac{\partial f}{\partial x} \quad \frac{\partial^2 f}{\partial x \partial y} \quad \frac{\partial^2 f}{\partial x^2} \quad \frac{\partial^3 f}{\partial x^2 \partial y} \quad \frac{\partial^3 f}{\partial x \partial y \partial x} \quad \frac{\partial}{\partial t} \quad \frac{\partial^2 u}{\partial x_1^2}$$

$$H_f = \begin{pmatrix} \dfrac{\partial^2 f(v)}{\partial x_1^2} & \dfrac{\partial^2 f(v)}{\partial x_1 \partial x_2} \\ \dfrac{\partial^2 f(v)}{\partial x_2 \partial x_1} & \dfrac{\partial^2 f(v)}{\partial x_2^2} \end{pmatrix}$$

$$H_f = \begin{pmatrix} \dfrac{\partial^2 f(v)}{\partial x_1^2} & \dfrac{\partial^2 f(v)}{\partial x_1 \partial x_2} & \dfrac{\partial^2 f(v)}{\partial x_1 \partial x_3} \\ \dfrac{\partial^2 f(v)}{\partial x_2 \partial x_1} & \dfrac{\partial^2 f(v)}{\partial x_2^2} & \dfrac{\partial^2 f(v)}{\partial x_2 \partial x_3} \\ \dfrac{\partial^2 f(v)}{\partial x_3 \partial x_1} & \dfrac{\partial^2 f(v)}{\partial x_3 \partial x_2} & \dfrac{\partial^2 f(v)}{\partial x_3^2} \end{pmatrix}$$

$$H_g = \begin{pmatrix} \dfrac{\partial^2 g(v)}{\partial y_1^2} & \dfrac{\partial^2 g(v)}{\partial y_1 \partial y_2} & \cdots & \dfrac{\partial^2 g(v)}{\partial y_1 \partial y_n} \\ \dfrac{\partial^2 g(v)}{\partial y_2 \partial y_1} & \dfrac{\partial^2 g(v)}{\partial y_2^2} & \cdots & \dfrac{\partial^2 g(v)}{\partial y_2 \partial y_n} \\ \vdots & \vdots & \ddots & \vdots \\ \dfrac{\partial^2 g(v)}{\partial y_n \partial y_1} & \dfrac{\partial^2 g(v)}{\partial y_n \partial y_2} & \cdots & \dfrac{\partial^2 g(v)}{\partial y_n^2} \end{pmatrix}$$

$$H_f(\mathbf{a}) = \begin{pmatrix} \dfrac{\partial^2 f(\mathbf{a})}{\partial x_1^2} & \dfrac{\partial^2 f(\mathbf{a})}{\partial x_1 \partial x_2} \\ \dfrac{\partial^2 f(\mathbf{a})}{\partial x_2 \partial x_1} & \dfrac{\partial^2 f(\mathbf{a})}{\partial x_2^2} \end{pmatrix} \qquad H_f = \begin{pmatrix} \dfrac{\partial^2 f}{\partial x_1^2} & \dfrac{\partial^2 f}{\partial x_1 \partial x_2} \\ \dfrac{\partial^2 f}{\partial x_2 \partial x_1} & \dfrac{\partial^2 f}{\partial x_2^2} \end{pmatrix}$$

$$I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad I_4 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad I_n = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix}$$

Proposition 1 (Identity)**.** *For every $n \times n$ matrix A we have $I_n A = A$.*

Exercise 2. Compute $\begin{pmatrix} \dfrac{\partial^2 f}{\partial x_1^2} & \dfrac{\partial^2 f}{\partial x_1 \partial x_2} \\ \dfrac{\partial^2 f}{\partial x_2 \partial x_1} & \dfrac{\partial^2 f}{\partial x_2^2} \end{pmatrix}$ for $f(x_1, x_2) = x_1^2 x_2$.

By **Proposition 1**, **Exercise 2** needs no identity matrix. See also **Theorem 4.2** in the course notes.

Added from IMO26 notation

$$\frac{\mathrm{d}f}{\mathrm{d}x} \quad \frac{\mathrm{d}^2 f}{\mathrm{d}x^2} \quad \frac{\mathrm{d}}{\mathrm{d}t}$$

$$\nabla f(v) = \begin{pmatrix} \frac{\partial f(v)}{\partial x_1} & \frac{\partial f(v)}{\partial x_2} & \frac{\partial f(v)}{\partial x_3} \end{pmatrix} \qquad \nabla g = \begin{pmatrix} \frac{\partial g}{\partial y_1} & \frac{\partial g}{\partial y_2} & \cdots & \frac{\partial g}{\partial y_n} \end{pmatrix}$$

$$f'(v) = \begin{pmatrix} \frac{\partial f_1(v)}{\partial x_1} & \frac{\partial f_1(v)}{\partial x_2} & \frac{\partial f_1(v)}{\partial x_3} \\ \frac{\partial f_2(v)}{\partial x_1} & \frac{\partial f_2(v)}{\partial x_2} & \frac{\partial f_2(v)}{\partial x_3} \end{pmatrix} \qquad f' = \begin{pmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \cdots & \frac{\partial f_2}{\partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1} & \frac{\partial f_m}{\partial x_2} & \cdots & \frac{\partial f_m}{\partial x_n} \end{pmatrix}$$

$$B(v, \epsilon) = \{u \in \mathbb{R}^d \mid d(u, v) < \epsilon\}, \quad S^\circ, \quad \partial S, \quad x^2 - 5x + 6 \in \mathbb{R}[x], \quad \mathbb{R}[t]$$

$$\left(x, y\right), \quad \left(1, 2\right), \quad \begin{pmatrix} x \\ y \\ z \end{pmatrix}, \quad \begin{pmatrix} 1 \\ 2 \end{pmatrix}^T, \quad \left(a\right)$$

$$x = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \in \mathbb{R}^n, \quad \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix}, \quad \begin{pmatrix} w_1 & w_2 & w_3 & w_4 \end{pmatrix}, \quad \begin{pmatrix} x_1 & x_2 & \cdots & x_n \end{pmatrix}$$

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}, \quad B = \begin{pmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \end{pmatrix}, \quad A^T$$

$$\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R}$$

$$\|x\|, \quad |x - y|, \quad \left\|\frac{u}{2}\right\|, \quad \langle u, v \rangle = u^T v$$

$$\{1, 2, 3\}, \quad \{x \in S \mid p(x)\}, \quad \left\{ x \in \mathbb{R}^n \;\middle|\; \|x\| \leq \frac{1}{2} \right\}$$

$$f \colon \mathbb{R}^n \to \mathbb{R}, \quad x - y \coloneqq x + (-y)$$

$$x^* = \arg\min_{x \in C} f(x), \qquad \min_{\substack{x \in \mathbb{R}^n \\ Ax \leq b}} f(x) \qquad \max_{x \geq 0} c^T x$$

$$H_f = \begin{pmatrix} \frac{\partial^2 f(v)}{\partial x_1^2} & \frac{\partial^2 f(v)}{\partial x_1 \partial x_2} \\ \frac{\partial^2 f(v)}{\partial x_2 \partial x_1} & \frac{\partial^2 f(v)}{\partial x_2^2} \end{pmatrix}, \quad I_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$S = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{12} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1n} & a_{2n} & \cdots & a_{nn} \end{pmatrix}, \quad \begin{pmatrix} s_{11} & s_{12} & s_{13} \\ s_{12} & s_{22} & s_{23} \\ s_{13} & s_{23} & s_{33} \end{pmatrix}, \quad \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{pmatrix}, \quad \begin{pmatrix} a & b \\ b & c \end{pmatrix}$$

$$\begin{pmatrix} e & 0 \\ 0 & d \end{pmatrix}, \quad \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{pmatrix}, \quad \begin{pmatrix} e & 0 \\ 0 & d \end{pmatrix}, \quad \begin{pmatrix} d_1 & 0 & \cdots & 0 \\ 0 & d_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & d_n \end{pmatrix}, \quad \begin{pmatrix} e & 0 \\ 0 & d \end{pmatrix}, \quad \begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{pmatrix}$$

Bracket style

Default (round):

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad \begin{pmatrix} e & 0 \\ 0 & d \end{pmatrix}$$

After `\usematb`:

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \quad \begin{bmatrix} \dfrac{\partial f(v)}{\partial x_1} & \dfrac{\partial f(v)}{\partial x_2} \end{bmatrix}$$

After `\usematp`:

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Code and pseudocode

Listing 1: Newton's method

```python
def newton(f, df, x0, tol=1e-10, max_iter=50):
    """Find a root of f with Newton's method."""
    x = x0
    for _ in range(max_iter):
        step = f(x) / df(x)
        x -= step
        if abs(step) < tol:  # converged
            return x
    raise RuntimeError("no convergence")


print(newton(lambda x: x**2 - 2, lambda x: 2 * x, 1.0))
```

Inline: `x -= f(x) / df(x)`.

Listing 2: Pseudocode from a plain .txt file

```
function hello_world(x, y)
    if (x ≥ y) then
```

```
        print(y)
        return x
    else if (x < y) then
        print(y)
        return y
    end
    for i = 1 to n do
        s ← s + a_i
    end for
```